Humana. Claims Payment Policy

Subject: Laboratory – Venous and Arterial Thrombosis Testing

Application: Medicare Advantage Products
Policy number: LAB2023006
Published date: 03/2023
Revision date: 12/2025

Disclaimer**: The intended audience of this medical claims payment policy is healthcare providers who treat Humana members. This policy is made available to provide information on certain Humana claims payment policies. This policy is a guideline only and does not constitute medical advice, guarantee of payment, plan pre-authorization, an explanation of benefits, or a contract. This policy does not govern whether a procedure is covered under a specific member plan or policy, nor is it intended to address every claim situation. Claims may be affected by other factors, such as: state and federal laws and regulations, provider contract terms, and our professional judgment. This policy is subject to change or termination by Humana. Humana has full and final discretionary authority for its interpretation and application. No part of this policy may be reproduced, stored in a retrieval system or transmitted, in any shape or form or by any means, electronic, mechanical, photocopying or otherwise without express written permission from Humana. When printed, this version becomes uncontrolled. For the most current information, always refer to the following website: **Humana.com/ClaimPaymentPolicies

Medicare Advantage Payment Policy

As a Medicare Advantage plan, we have developed payment policies to explain coverage more fully for services. The criteria outlined will apply to Medicare Advantage plans except when the procedure(s) has established criteria through the Original Medicare National Coverage Determinations (NCDs) and Local Coverage Determinations (LCDs) as outlined below.

In addition to this policy, claims payments are subject to other plan requirements for the processing and payment of claims, including, but not limited to, requirements of medical necessity and reasonableness and applicable referral or authorization requirements.

- Plasma testing for protein C deficiency (CPT code 85302, CPT code 85303), protein S deficiency (CPT code 85305, CPT code 85306) and antithrombin III deficiency (CPT code 85300, CPT code 85301) **IS REIMBURSABLE** for any of the following situations:
  - Acute coronary syndromes including myocardial infarction
  - Any deep venous thrombosis
  - Arterial ischemic stroke
  - Deep phlebothrombosis in pregnancy
  - Disseminated intravascular coagulation, also known as defibrination syndrome
  - Pulmonary embolism
  - Relatives of individuals with deep venous thrombosis

- Plasma testing for protein C deficiency (CPT code 85302, CPT code 85303) or protein S deficiency (CPT code 85305, CPT code 85306) **IS REIMBURSABLE** for any of the following situations:
  - Neonatal purpura fulminans
  - Warfarin-induced skin necrosis

- Venous thrombosis risk testing (CPT codes 85300-85303, CPT code 85305, CPT code 85306) for superficial venous thrombosis, including superficial thrombophlebitis and varicosities, **IS NOT REIMBURSABLE**.
- The activated protein C resistance assay (CPT code 85307) **IS NOT REIMBURSABLE**.
- Multi-gene panel testing (CPT code 0278U*) **IS NOT REIMBURSABLE**.

Applicable procedures for Medicare Advantage plans

Procedure Code	Code Description
85300	Clotting inhibitors or anticoagulants; antithrombin III, activity
85301	Clotting inhibitors or anticoagulants; antithrombin III, antigen assay
85302	Clotting inhibitors or anticoagulants; protein C, antigen
85303	Clotting inhibitors or anticoagulants; protein C, activity
85305	Clotting inhibitors or anticoagulants; protein S, total

Procedure Code	Code Description
85306	Clotting inhibitors or anticoagulants; protein S, free
85307	Activated Protein C (APC) resistance assay
0278U*	Hematology (genetic thrombosis), genomic sequence analysis of 14 genes, blood, buccal swab, or amniotic fluid Proprietary test: Versiti™ Thrombosis Panel Lab/Manufacturer: Versiti™ Comprehensive Bleeding Disorder Panel

Original Medicare Local Coverage Determinations (LCDs)

Contractor Name	Policy Title	Policy Number	Procedure Code
First Coast Service Options, Inc.	Billing and Coding: Molecular Pathology and Genetic Testing	A58918	0278U
Palmetto GBA	MolDX: Molecular Diagnostic Tests (MDT)	L35025	0278U
Novitas Solutions, Inc	Biomarkers Overview	L35062	0278U
Novitas Solutions, Inc	Genetic Testing in Oncology: Specific Tests	L39365	0278U
Noridian Healthcare Solutions, LLC	MolDX: Molecular Diagnostic Tests (MDT)	L35160	0278U
CGS Administrators, LLC	MolDX: Molecular Diagnostic Tests (MDT)	L36021	0278U
Noridian Healthcare Solutions, LLC	MolDX: Molecular Diagnostic Tests (MDT)	L36256	0278U
Wisconsin Physicians Service Insurance Corporation	MolDX: Molecular Diagnostic Tests (MDT)	L36807	0278U
CGS Administrators, LLC	MolDX: Repeat Germline Testing	L38288	0278U
Noridian Healthcare Solutions, LLC	MolDX: Repeat Germline Testing	L38351	0278U
Noridian Healthcare Solutions, LLC	MolDX: Repeat Germline Testing	L38353	0278U

Contractor Name	Policy Title	Policy Number	Procedure Code
Palmetto GBA	MolDX: Repeat Germline Testing	L38274	0278U
Wisconsin Physicians Service Insurance Corporation	MolDX: Repeat Germline Testing	L38429	0278U

*The payment criteria for 0278U apply to all Medicare jurisdictions, except those states awarded to CGS Administrators, First Coast Service Options, Noridian Healthcare Solutions, Novitas Solutions, Palmetto GBA, and Wisconsin Physicians Service Insurance Company.

References

ACOG. (2013). ACOG Practice Bulletin No. 138: Inherited thrombophilias in pregnancy. *Obstet Gynecol, 122*(3), 706-717. doi:10.1097/01.AOG.0000433981.36184.4e

ACOG. (2018). ACOG Practice Bulletin No. 197 Summary: Inherited Thrombophilias in Pregnancy. *Obstet Gynecol, 132*(1), 249-251. doi:10.1097/aog.0000000000002705

Algahtani, F. H., & Stuckey, R. (2019). High factor VIII levels and arterial thrombosis: illustrative case and literature review. *Ther Adv Hematol, 10*, 2040620719886685. doi:10.1177/2040620719886685

ASCLS. (2021). American Society for Clinical Laboratory Science. Retrieved from https://www.choosingwisely.org/clinician-lists/ascls7-do-not-order-a-homocysteine-assay-as-part-of-the-thrombophilia-work-up/

ASCP. (2017). American Society for Clinical Pathology. Retrieved from http://www.choosingwisely.org/clinician-lists/ascp-testing-for-protein-c-protein-s-or-antithrombin-during-active-clotting-event/

ASCP. (2019). American Society of Clinical Pathology. Retrieved from https://www.choosingwisely.org/clinician-lists/ascp-hypercoagulable-workup/

ASH. (2013). ASH - Testing for thromboembolism | Choosing Wisely. Retrieved from http://www.choosingwisely.org/clinician-lists/american-society-hematology-testing-for-thrombophilia-in-adults/. http://www.choosingwisely.org/clinician-lists/american-society-hematology-testing-for-thrombophilia-in-adults/

Bank, I., Libourel, E. J., Middeldorp, S., Hamulyak, K., van Pampus, E. C., Koopman, M. M., . . . Buller, H. R. (2005). Elevated levels of FVIII:C within families are associated with an increased risk for venous and arterial thrombosis. *J Thromb Haemost, 3*(1), 79-84. doi:10.1111/j.1538-7836.2004.01033.x

Barnes, G. (2017, 06/05/2017). Thrombophilia Testing for Provoked VTE. Retrieved from https://www.acc.org/latest-in-cardiology/ten-points-to-remember/2017/06/05/12/46/thrombophilia-testing-in-provoked-venous-thromboembolism

Barnes, G. D. (2017). Thrombophilia Testing and Venous Thrombosis. Retrieved from https://www.acc.org/latest-in-cardiology/ten-points-to-remember/2017/10/20/11/18/thrombophilia-testing-and-venous-thrombosis.

Bartholomew, J. R. (2017). Update on the management of venous thromboembolism. *Cleve Clin J Med, 84*(12 Suppl 3), 39-46. doi:10.3949/ccjm.84.s3.04

Bashford, M. T., Hickey, S. E., Curry, C. J., Toriello, H. V., American College of Medical, G., Genomics Professional, P., & Guidelines, C. (2020). Addendum: ACMG Practice Guideline: lack of evidence for MTHFR polymorphism testing. *Genetics in Medicine, 22*(12), 2125-2125. doi:10.1038/s41436-020-0843-0

Bauer, K. (2019a). Factor V Leiden and activated protein C resistance. Retrieved from https://www.uptodate.com/contents/factor-v-leiden-and-activated-protein-c-resistance?topicRef=1361&source=see_link

Bauer, K. (2019b). Protein C deficiency. Retrieved from https://www.uptodate.com/contents/protein-c-deficiency?topicRef=1361&source=see_link

Bauer, K. (2019c). Prothrombin G20210A mutation. Retrieved from https://www.uptodate.com/contents/prothrombin-g20210a-mutation?topicRef=1361&source=see_link

Bauer, K. (2020). Clinical presentation and diagnosis of the nonpregnant adult with suspected deep vein thrombosis of the lower extremity. Retrieved from https://www.uptodate.com/contents/clinical-presentation-and-diagnosis-of-the-nonpregnant-adult-with-suspected-deep-vein-thrombosis-of-the-lower-

extremity?search=deep%20vein%20thrombosis&source=search_result&selectedTitle=2~150&usage_type=default&display_rank=2

Bauer, K. (2021). Protein S deficiency. Retrieved from https://www.uptodate.com/contents/protein-s-deficiency?topicRef=1361&source=see_link

Bauer, K., & Lip, G. (2018). Overview of the causes of venous thrombosis - UpToDate. In G. Finlay (Ed.), *UpToDate*. Retrieved from https://www.uptodate.com/contents/overview-of-the-causes-of-venous-thrombosis?search=thrombophilia&usage_type=default&source=search_result&selectedTitle=1~150&display_rank=1

Bauer, K., & Lip, G. (2020). Overview of the causes of venous thrombosis - UpToDate. In G. Finlay (Ed.), *UpToDate*. Retrieved from https://www.uptodate.com/contents/overview-of-the-causes-of-venous-thrombosis?search=thrombophilia&usage_type=default&source=search_result&selectedTitle=1~150&display_rank=1

Byrnes, J. R., & Wolberg, A. S. (2017). Red blood cells in thrombosis. *Blood, 130*(16), 1795-1799. doi:10.1182/blood-2017-03-745349

Carroll, B. J., & Piazza, G. (2018). Hypercoagulable states in arterial and venous thrombosis: When, how, and who to test? *Vasc Med, 23*(4), 388-399. doi:10.1177/1358863x18755927

Chiasakul, T., De Jesus, E., Tong, J., Chen, Y., Crowther, M., Garcia, D., . . . Cuker, A. (2019). Inherited Thrombophilia and the Risk of Arterial Ischemic Stroke: A Systematic Review and Meta-Analysis. *J Am Heart Assoc, 8*(19), e012877. doi:10.1161/jaha.119.012877

Connors, J. M. (2017). Thrombophilia Testing and Venous Thrombosis. In *N Engl J Med* (Vol. 377, pp. 2298). United States.

Crous-Bou, M., Harrington, L. B., & Kabrhel, C. (2016). Environmental and genetic risk factors associated with venous thromboembolism. *Semin Thromb Hemost, 42*(8), 808-820. doi:10.1055/s-0036-1592333

Curtis, C., Mineyko, A., Massicotte, P., Leaker, M., Jiang, X. Y., Floer, A., & Kirton, A. (2017). Thrombophilia risk is not increased in children after perinatal stroke. *Blood, 129*(20), 2793-2800. doi:10.1182/blood-2016-11-750893

EGAPP. (2011). Recommendations from the EGAPP Working Group: routine testing for Factor V Leiden (R506Q) and prothrombin (20210G>A) mutations in adults with a history of idiopathic venous thromboembolism and their adult family members. *Genet Med, 13*(1), 67-76. doi:10.1097/GIM.0b013e3181fbe46f

Ferriero, D. M., , Fullerton, H. J., , Bernard, T. J., Billinghurst, L., . . . Nursing, A. H. A. S. C. a. C. o. C. a. S. (2019). Management of Stroke in Neonates and Children: A Scientific Statement From the American Heart Association/American Stroke Association. *Stroke, 50*(3). doi: http://dx.doi.org/10.1161/str.0000000000000183

Gupta, A., Sarode, R., & Nagalla, S. (2017). Thrombophilia Testing in Provoked Venous Thromboembolism: A Teachable Moment. *JAMA Internal Medicine, 177*(8), 1195-1196. doi:10.1001/jamainternmed.2017.1815

Herrmann, J. (2018). *Clinical Cardio-Oncology*: Elsevier.

Hickey, S. E., Curry, C. J., & Toriello, H. V. (2013). ACMG Practice Guideline: lack of evidence for MTHFR polymorphism testing. *Genet Med, 15*(2), 153-156. doi:10.1038/gim.2012.165

Kleindorfer, D. O., Towfighi, A., Chaturvedi, S., Cockroft, K. M., Gutierrez, J., Lombardi-Hill, D., . . . Williams, L. S. (2021). 2021 Guideline for the Prevention of Stroke in Patients With Stroke and Transient Ischemic Attack: A Guideline From the American Heart Association/American Stroke Association. *Stroke, 52*(7), e364-e467. doi:10.1161/str.0000000000000375

Konstantinides, S. V., Meyer, G., Becattini, C., Bueno, H., Geersing, G. J., Harjola, V. P., . . . Zamorano, J. L. (2019). 2019 ESC Guidelines for the diagnosis and management of acute pulmonary embolism developed in collaboration with the European Respiratory Society (ERS): The Task Force for the diagnosis and management of acute pulmonary embolism of the European Society of Cardiology (ESC). *Eur Respir J, 54*(3). doi:10.1183/13993003.01647-2019

Kujovich, J. L. (2011). Factor V Leiden thrombophilia. *Genet Med, 13*(1), 1-16. doi:10.1097/GIM.0b013e3181faa0f2

Kujovich, J. L. (2018). Factor V Leiden Thrombophilia. In M. P. Adam, H. H. Ardinger, R. A. Pagon, S. E. Wallace, L. J. H. Bean, K. Stephens, & A. Amemiya (Eds.), *GeneReviews((R))*. Seattle (WA): University of Washington, Seattle

University of Washington, Seattle. GeneReviews is a registered trademark of the University of Washington, Seattle. All rights reserved.

Lee, E. J., Dykas, D. J., Leavitt, A. D., Camire, R. M., Ebberink, E., García de Frutos, P., . . . Lee, A. I. (2017). Whole-exome sequencing in evaluation of patients with venous thromboembolism. *Blood Adv, 1*(16), 1224-1237. doi:10.1182/bloodadvances.2017005249

Lehman, L. L., Beaute, J., Kapur, K., Danehy, A. R., Bernson-Leung, M. E., Malkin, H., . . . Trenor, C. C. (2017). Workup for Perinatal Stroke Does Not Predict Recurrence. *Stroke, 48*(8), 2078-2083. doi:10.1161/STROKEAHA.117.017356

Lim, W., Le Gal, G., Bates, S. M., Righini, M., Haramati, L. B., Lang, E., . . . Mustafa, R. A. (2018). American Society of Hematology 2018 guidelines for management of venous thromboembolism: diagnosis of venous thromboembolism. *Blood Adv, 2*(22), 3226. doi:10.1182/bloodadvances.2018024828

Linkins, L. A., & Takach Lapner, S. (2017). Review of D-dimer testing: Good, Bad, and Ugly. *Int J Lab Hematol, 39 Suppl 1*, 98-103. doi:10.1111/ijlh.12665

Mazzolai, L., Ageno, W., Alatri, A., Bauersachs, R., Becattini, C., Brodmann, M., . . . Aboyans, V. (2021). Second consensus document on diagnosis and management of acute deep vein thrombosis: updated document elaborated by the ESC Working Group on aorta and peripheral vascular diseases and the ESC Working Group on pulmonary circulation and right ventricular function. *European Journal of Preventive Cardiology*. doi:10.1093/eurjpc/zwab088

Murphy, C. H., & Sabath, D. E. (2019). Comparison of Phenotypic Activated Protein C Resistance Testing With a Genetic Assay for Factor V Leiden. *Am J Clin Pathol, 151*(3), 302-305. doi:10.1093/ajcp/aqy142

NATF. (2019). Genetic Risk Factors for Blood Clots and the Role of Genetic Testing. Retrieved from https://natfonline.org/2019/01/genetic-risk-factors-blood-clots-role-genetic-testing/

Onda, S., Furukawa, K., Haruki, K., Hamura, R., Shirai, Y., Yasuda, J., . . . Ikegami, T. (2021). d-dimer-based screening for early diagnosis of venous thromboembolism after hepatectomy. *Langenbeck's Archives of Surgery, 406*(3), 883-892. doi:10.1007/s00423-020-02058-9

Ordieres-Ortega, L., Demelo-Rodríguez, P., Galeano-Valle, F., Kremers, B. M. M., ten Cate-Hoek, A. J., & ten Cate, H. (2020). Predictive value of D-dimer testing for the diagnosis of venous thrombosis in unusual locations: A systematic review. *Thrombosis Research, 189*, 5-12. doi:https://doi.org/10.1016/j.thromres.2020.02.009

Ortel, T. L., Neumann, I., Ageno, W., Beyth, R., Clark, N. P., Cuker, A., . . . Zhang, Y. (2020). American Society of Hematology 2020 guidelines for management of venous thromboembolism: treatment of deep vein thrombosis and pulmonary embolism. *Blood Adv, 4*(19), 4693-4738. doi:10.1182/bloodadvances.2020001830

Previtali, E., Bucciarelli, P., Passamonti, S. M., & Martinelli, I. (2011). Risk factors for venous and arterial thrombosis. *Blood Transfus, 9*(2), 120-138. doi:10.2450/2010.0066-10

Raffini, L., Mahoney, D. H., & Armsby, C. (2019). Thrombophilia testing in children and adolescents. *UpToDate*. Retrieved from https://www.uptodate.com/contents/thrombophilia-testing-in-children-and-adolescents?topicRef=1354&source=see_link#H1628041305

Segal, J. B., Brotman, D. J., Emadi, A., Necochea, A. J., Samal, L., Wilson, L. M., . . . Bass, E. B. (2009). Outcomes of genetic testing in adults with a history of venous thromboembolism. *Evid Rep Technol Assess (Full Rep)*(180), 1-162.

SIGN. (2014). Prevention and management of venous thromboembolism Retrieved from https://www.sign.ac.uk/assets/sign122.pdf

Stevens, S. M., Woller, S. C., Bauer, K. A., Kasthuri, R., Cushman, M., Streiff, M., . . . Douketis, J. D. (2016). Guidance for the evaluation and treatment of hereditary and acquired thrombophilia. *J Thromb Thrombolysis, 41*, 154-164. doi:10.1007/s11239-015-1316-1

SVM. (2013, 02/21/2013). Don't do work up for clotting disorder (order hypercoagulable testing) for patients who develop first episode of deep vein thrombosis (DVT) in the setting of a known cause. Retrieved from http://www.choosingwisely.org/clinician-lists/society-vascular-medicine-clotting-disorder-workup-after-first-episode-of-deep-vein-thrombosis/

Thompson, B. T., Kabrhel, Christopher. (2018). Overview of acute pulmonary embolism in adults. Retrieved from https://www.uptodate.com/contents/overview-of-acute-pulmonary-embolism-in-adults?search=pulmonary%20embolism&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1

Thompson, B. T., Kabrhel, Christopher. (2020). Overview of acute pulmonary embolism in adults. Retrieved from https://www.uptodate.com/contents/overview-of-acute-pulmonary-embolism-in-adults?search=pulmonary%20embolism&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1

Zhang, S., Taylor, A. K., Huang, X., Luo, B., Spector, E. B., Fang, P., & Richards, C. S. (2018). Venous thromboembolism laboratory testing (factor V Leiden and factor II c.*97G>A), 2018 update: a technical standard of the American College of Medical Genetics and Genomics (ACMG). *Genet Med, 20*(12), 1489-1498. doi:10.1038/s41436-018-0322-z